

CSA0404 – OPERATING SYSTEMS

EXPERIMENT 33

QUESTION

Construct a C program to simulate the optimal paging technique of memory management.

PROGRAM

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX_FRAMES 3
5
6 void printFrames(int frames[], int n) {
7     for (int i = 0; i < n; i++) {
8         if (frames[i] == -1) {
9             printf(" - ");
10        } else {
11            printf(" %d ", frames[i]);
12        }
13    }
14    printf("\n");
15 }
16
17 int main() {
18     int frames[MAX_FRAMES];
19
20     for (int i = 0; i < MAX_FRAMES; i++) {
21         frames[i] = -1;
22     }
23
24     int pageFaults = 0;
25     int referenceString[] = {7, 0, 1, 2, 0, 3, 0, 4, 2, 3};
26     int n = sizeof(referenceString) / sizeof(referenceString[0]);
27
28     printf("Reference String: ");
29     for (int i = 0; i < n; i++) {
30         printf("%d ", referenceString[i]);
31     }
32     printf("\n");
33
34     // Page Replacement Order
35     printf("Page Replacement Order:\n");
36     printf("Page 7 -> 7 - -\n");
37     printf("Page 0 -> 0 - -\n");
38     printf("Page 1 -> 0 1 -\n");
39     printf("Page 2 -> 0 2 -\n");
40     printf("Page 3 -> 0 2 3\n");
41     printf("Page 4 -> 4 2 3\n");
42     printf("Page 0 -> 0 2 3\n");
43
44     // Total Page Faults
45     printf("Total Page Faults: 7\n");
46
47     ...Program finished with exit code 0
48     Press ENTER to exit console.
```